Collections

1. Create an ArrayList of type String with 10 string elements. Add 10 string elements to ArrayList and perform the below operations
 Add an element to the ArrayList
 Iterate through the ArrayList by using Iterator object
 Add an element at a specific index
 Remove an element from the ArrayList, Remove at an index
 Update the element at a specific index
 Check the element is present at a particular index
 Get an element at a particular index
 Find out the size of the ArrayList
 Check the given element is present in the ArrayList
 Remove all elements of the ArrayList

 Answer:

```
import java.util.*;
public class ArrayListDemo {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();

        // Add 10 elements
        list.add("Java");
        list.add("Python");
        list.add("C");
        list.add("C++");
        list.add("HTML");
        list.add("CSS");
        list.add("JavaScript");
        list.add("SQL");
        list.add("Spring");
        list.add("Angular");

        // Add an element
        list.add("React");

        // Iterate using Iterator
        System.out.println("ArrayList Elements:");
        Iterator<String> itr = list.iterator();
        while (itr.hasNext()) {
```

```
            System.out.println(itr.next());
        }

        // Add element at specific index
        list.add(2, "PHP");

        // Remove element by value
        list.remove("CSS");

        // Remove element by index
        list.remove(3);

        // Update element
        list.set(0, "Java Core");

        // Check element at index
        System.out.println("Element at index 2: " +
list.get(2));

        // Get element at index
        System.out.println("Element at index 4: " +
list.get(4));

        // Size
        System.out.println("Size: " + list.size());

        // Check contains
        System.out.println("Contains SQL? " +
list.contains("SQL"));

        // Remove all elements
        list.clear();

        System.out.println("After clear: " + list);
    }
}
```

2. Create a HashMap with at least 10 key value pairs of the Student ID and Name
   Insert a Key value mapping into the map
   Fetch the value of a Key
   Create a clone/copy of HashMap
   Check if the given Key is in the Map

Check if the value is in the Map
Check if the map is empty
Print the size of the Map to the console
Print all the Keys of the map to the console
Print all the Keys of the map to the console
Remove a specific Key-value pair
Copy all the elements of the Map to another Map

Answer:

```
import java.util.*;
public class HashMapDemo {
    public static void main(String[] args) {
        HashMap<Integer, String> map = new HashMap<>();

        // Add 10 key-value pairs
        map.put(101, "Naveen");
        map.put(102, "Ravi");
        map.put(103, "Kiran");
        map.put(104, "Ajay");
        map.put(105, "Rahul");
        map.put(106, "Ramesh");
        map.put(107, "Suresh");
        map.put(108, "Anil");
        map.put(109, "Lokesh");
        map.put(110, "Vijay");

        // Insert new key-value
        map.put(111, "Manoj");

        // Fetch value
        System.out.println("Student 101: " +
map.get(101));

        // Clone map
        HashMap<Integer, String> copy =
                (HashMap<Integer, String>) map.clone();

        System.out.println("Cloned Map: " + copy);

        // Check key
        System.out.println(map.containsKey(105));
```

```
        // Check value
        System.out.println(map.containsValue("Ravi"));

        // Is Empty
        System.out.println(map.isEmpty());

        // Size
        System.out.println("Size: " + map.size());

        // Print keys
        System.out.println("Keys:");
        System.out.println(map.keySet());

        // Print values
        System.out.println("Values:");
        System.out.println(map.values());

        // Remove key-value
        map.remove(110);

        // Copy to another map
        HashMap<Integer, String> map2 = new HashMap<>();
        map2.putAll(map);

        System.out.println("Copied Map:");
        System.out.println(map2);
    }
}
```

3. Create a HashSet with at least 10 elements of type String Write program covering all the operations of HashSet

Answer:

```
import java.util.*;

public class HashSetDemo {

    public static void main(String[] args) {

        HashSet<String> set = new HashSet<>();

        // Add 10 elements

        set.add("Java");

        set.add("Python");
```

```
        set.add("C");
        set.add("C++");
        set.add("HTML");
        set.add("CSS");
        set.add("SQL");
        set.add("Spring");
        set.add("Angular");
        set.add("React");

        // Display
        System.out.println(set);

        // Add duplicate
        set.add("Java");

        // Iterate
        System.out.println("Using Iterator:");
        Iterator<String> itr = set.iterator();
        while (itr.hasNext()) {
            System.out.println(itr.next());
        }

        // Contains
        System.out.println("Contains SQL? " +
set.contains("SQL"));

        // Remove element
        set.remove("HTML");

        // Size
```

```
        System.out.println("Size: " + set.size());

        // Is Empty
        System.out.println("Is Empty? " + set.isEmpty());

        // Clone
        HashSet<String> copy =
                (HashSet<String>) set.clone();

        System.out.println("Cloned Set:");
        System.out.println(copy);

        // Clear
        set.clear();

        System.out.println("After Clear:");
        System.out.println(set);
    }
}
```